Great Place To Work. Certified FEB 2023 – FEB 2024 INDIA

UPES

UNIVERSITY OF TOMORROW

nirf RANKING 2023

#RANKED 52 IN INDIA

#University Category

QS WORLD UNIVERSITY RANKINGS

NO.1 PVT. UNIVERSITY IN ACADEMIC REPUTATION IN INDIA

NAAC

ACCREDITED GRADE 'A' BY NAAC

PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 01 – Lecture 03: AI Techniques, Limitations & Impact

Tofik Ali

School of Computer Science, UPES Dehradun

February 28, 2026

Topic focus: **what AI can do**, where it fails, and why it matters

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.

Quick Links

Techniques Limitations Impact Case Summary

Agenda

Techniques recap

Limitations

Impact and domains

Mini case

Summary

Learning Outcomes

By the end of this lecture, you should be able to:

- Recall core AI technique families and typical use-cases.

Learning Outcomes

By the end of this lecture, you should be able to:

- Recall core AI technique families and typical use-cases.
- Explain major limitations (data, robustness, bias, explainability).

Learning Outcomes

By the end of this lecture, you should be able to:

- Recall core AI technique families and typical use-cases.
- Explain major limitations (data, robustness, bias, explainability).
- Discuss the impact of AI on society and key application domains.

Abbreviations (Used in This Lecture)

- **OOD**: out-of-distribution (data unlike training data)

Abbreviations (Used in This Lecture)

- **OOD**: out-of-distribution (data unlike training data)
- **Drift**: changing patterns over time (a type of shift)

Abbreviations (Used in This Lecture)

- **OOD**: out-of-distribution (data unlike training data)
- **Drift**: changing patterns over time (a type of shift)
- **PII**: personally identifiable information (privacy)

Abbreviations (Used in This Lecture)

- **OOD**: out-of-distribution (data unlike training data)
- **Drift**: changing patterns over time (a type of shift)
- **PII**: personally identifiable information (privacy)
- **KPI**: key performance indicator (business/operational metric)

Technique Families (Big Picture)

- **Search and optimization**: explore possible actions/solutions (planning, games).

Technique Families (Big Picture)

- **Search and optimization**: explore possible actions/solutions (planning, games).
- **Knowledge and logic**: facts + rules + inference (Unit II).

Technique Families (Big Picture)

- **Search and optimization**: explore possible actions/solutions (planning, games).
- **Knowledge and logic**: facts + rules + inference (Unit II).
- **Machine learning**: learn patterns from data (Units III–IV).

Technique Families (Big Picture)

- **Search and optimization**: explore possible actions/solutions (planning, games).
- **Knowledge and logic**: facts + rules + inference (Unit II).
- **Machine learning**: learn patterns from data (Units III–IV).
- **Perception**: vision/speech/NLP (beyond this course, but important).

Why AI Systems Fail (Common Reasons)

- **Data limitations**: missing values, bias, low coverage, noisy labels

Why AI Systems Fail (Common Reasons)

- **Data limitations**: missing values, bias, low coverage, noisy labels
- **Distribution shift**: real world differs from training data

Why AI Systems Fail (Common Reasons)

- **Data limitations**: missing values, bias, low coverage, noisy labels
- **Distribution shift**: real world differs from training data
- **Robustness**: small changes cause large errors (adversarial/noise)

Why AI Systems Fail (Common Reasons)

- **Data limitations**: missing values, bias, low coverage, noisy labels
- **Distribution shift**: real world differs from training data
- **Robustness**: small changes cause large errors (adversarial/noise)
- **Explainability**: hard to justify decisions in high-stakes settings

Why AI Systems Fail (Common Reasons)

- **Data limitations**: missing values, bias, low coverage, noisy labels
- **Distribution shift**: real world differs from training data
- **Robustness**: small changes cause large errors (adversarial/noise)
- **Explainability**: hard to justify decisions in high-stakes settings
- **Safety & security**: harmful outputs, model misuse, privacy leakage

Distribution Shift: Common Types (Intuition)

- **Covariate shift**: input distribution changes ($P(X)$ changes).

Practical effect: your model can silently degrade even if the code is unchanged.

Distribution Shift: Common Types (Intuition)

- **Covariate shift**: input distribution changes ($P(X)$ changes).
- **Label shift**: class proportions change ($P(Y)$ changes).

Practical effect: your model can silently degrade even if the code is unchanged.

Distribution Shift: Common Types (Intuition)

- **Covariate shift**: input distribution changes ($P(X)$ changes).
- **Label shift**: class proportions change ($P(Y)$ changes).
- **Concept drift**: relationship changes ($P(Y|X)$ changes).

Practical effect: your model can silently degrade even if the code is unchanged.

Robustness vs Security (Short)

- **Robustness**: handle natural noise/variation (lighting, typos, sensor noise).

Both require testing beyond a single static test set.

Robustness vs Security (Short)

- **Robustness**: handle natural noise/variation (lighting, typos, sensor noise).
- **Security**: handle intentional attacks/misuse (adversarial inputs, fraudsters).

Both require testing beyond a single static test set.

A Simple Deployment Checklist

1. What is the **goal** and what are the real costs of errors?

A Simple Deployment Checklist

1. What is the **goal** and what are the real costs of errors?
2. Is the training data **representative** of deployment?

A Simple Deployment Checklist

1. What is the **goal** and what are the real costs of errors?
2. Is the training data **representative** of deployment?
3. Are we tracking **fairness** and **privacy** constraints?

A Simple Deployment Checklist

1. What is the **goal** and what are the real costs of errors?
2. Is the training data **representative** of deployment?
3. Are we tracking **fairness** and **privacy** constraints?
4. Do we have **monitoring** and retraining triggers (drift)?

What to Monitor After Deployment

- **Data drift signals**: feature distributions, missing values, outlier rates.

What to Monitor After Deployment

- **Data drift signals**: feature distributions, missing values, outlier rates.
- **Model signals**: prediction score distribution, confidence, calibration.

What to Monitor After Deployment

- **Data drift signals**: feature distributions, missing values, outlier rates.
- **Model signals**: prediction score distribution, confidence, calibration.
- **Outcome signals**: error rates when labels arrive; subgroup gaps (fairness).

What to Monitor After Deployment

- **Data drift signals**: feature distributions, missing values, outlier rates.
- **Model signals**: prediction score distribution, confidence, calibration.
- **Outcome signals**: error rates when labels arrive; subgroup gaps (fairness).
- **Operational signals**: latency, cost, uptime, user feedback.

Impact of AI (Balanced View)

- Benefits: automation, personalization, decision support, new capabilities

Impact of AI (Balanced View)

- Benefits: automation, personalization, decision support, new capabilities
- Risks: bias amplification, surveillance/privacy loss, unsafe automation, job shifts

Impact of AI (Balanced View)

- Benefits: automation, personalization, decision support, new capabilities
- Risks: bias amplification, surveillance/privacy loss, unsafe automation, job shifts
- Responsible practice: measure, audit, monitor, and communicate limits

Application Domains (Examples)

- Finance (fraud, credit), healthcare (risk/imaging), education (personalization)

Application Domains (Examples)

- Finance (fraud, credit), healthcare (risk/imaging), education (personalization)
- Security (anomaly detection), retail (recommendation), logistics (forecasting/routing)

Application Domains (Examples)

- Finance (fraud, credit), healthcare (risk/imaging), education (personalization)
- Security (anomaly detection), retail (recommendation), logistics (forecasting/routing)
- Energy/utilities (demand forecasting), public services (resource allocation)

Mini Case: Credit Scoring Model

Suppose a model predicts whether to approve a loan.

- Quality: accuracy is not enough — false negatives/positives have different costs.

Mini Case: Credit Scoring Model

Suppose a model predicts whether to approve a loan.

- Quality: accuracy is not enough — false negatives/positives have different costs.
- Fairness: are outcomes similar across groups? are proxies used?

Mini Case: Credit Scoring Model

Suppose a model predicts whether to approve a loan.

- Quality: accuracy is not enough — false negatives/positives have different costs.
- Fairness: are outcomes similar across groups? are proxies used?
- Explainability: can we justify why an application was rejected?

Mini Case: Credit Scoring Model

Suppose a model predicts whether to approve a loan.

- Quality: accuracy is not enough — false negatives/positives have different costs.
- Fairness: are outcomes similar across groups? are proxies used?
- Explainability: can we justify why an application was rejected?
- Monitoring: economic changes can cause drift (model becomes outdated).

Mini Case: Recommendation Feedback Loop

A recommender system can change the data it later trains on:

- showing some items more → more clicks → even more exposure

Mitigation ideas: exploration, diversity constraints, and monitoring.

Mini Case: Recommendation Feedback Loop

A recommender system can change the data it later trains on:

- showing some items more → more clicks → even more exposure
- can reduce diversity and amplify bias/popularity

Mitigation ideas: exploration, diversity constraints, and monitoring.

Summary

- AI includes multiple technique families: search, logic, learning.
- Real limitations: data quality, shift, robustness, bias, explainability, safety.
- AI has large impact — use case-by-case evaluation and responsible practices.

Sources: Rich and Knight, *Artificial Intelligence*, McGraw Hill, 2017.;Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 1–2 (Introduction, Intelligent Agents).;Poole and Mackworth, *Artificial Intelligence: Foundations of Computational Agents* (2e), selected sections.

Exit Questions

1. List four reasons why AI systems fail in deployment.
2. Give two examples of distribution shift.
3. In a credit scoring system, why is explainability important?
4. Name five application domains of AI and one task in each.